SMART INDIA HACKATHON 202

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TITLE PAGE

- **Problem Statement ID –** 26121
- **Problem Statement Title:-** eRTMAC-NWIS (Nearby Wells Intelligence System): An AI-Powered Offset Well Knowledge and Decision Support Platform for Drilling Operations
- **Theme -** Smart Automation
- **PS Category –** Software
- **Team ID -** 53476
- **Team Name –** Last 6 Braincells

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NWIS: INTELLIGENCE LAYER

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THE PROBLEM :

- **Historical Drilling Knowledge** (WCRs, DDRs, PDFs, Databases)
- ↓ **Scattered Across Multiple Sources**
- ↓ **Engineers Manually Search Reports**
- ↓ **Delayed Decision Making**
- ↓ **Higher Operational Risk**

PROPOSED SOLUTION

1. Institutional Memory
Convert historical drilling reports into a searchable knowledge base.

2. Smart Offset Intelligence
Map and identify the most relevant nearby wells.

3. Live-Historical Correlation
Compare current telemetry with historical well behaviour.

4. Proactive Risk Alerts
Detect upcoming risks and provide warnings.

5. Mitigation Strategies
Surface proven mitigation strategies from past wells.

UNIQUE SELLING POINTS

1 Contextual Analogue Engine
Finds relevant wells based on drilling, geological and operational similarity, not just nearest wells.

2 Historical Risk Radar
Flags upcoming risks by matching the active well's depth with offset-well risk history.

3 Explainable Alerts
Every alert is backed by historical wells, events and live telemetry showing why the risk is likely.

4 Mitigation Memory
Links past incidents with actions taken and outcomes to recommend what works.

5 System-Agnostic Layer
Seamlessly integrates with eRTMAC, WITSML, APIs and other real-time drilling systems.

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TECHNICAL APPROACH

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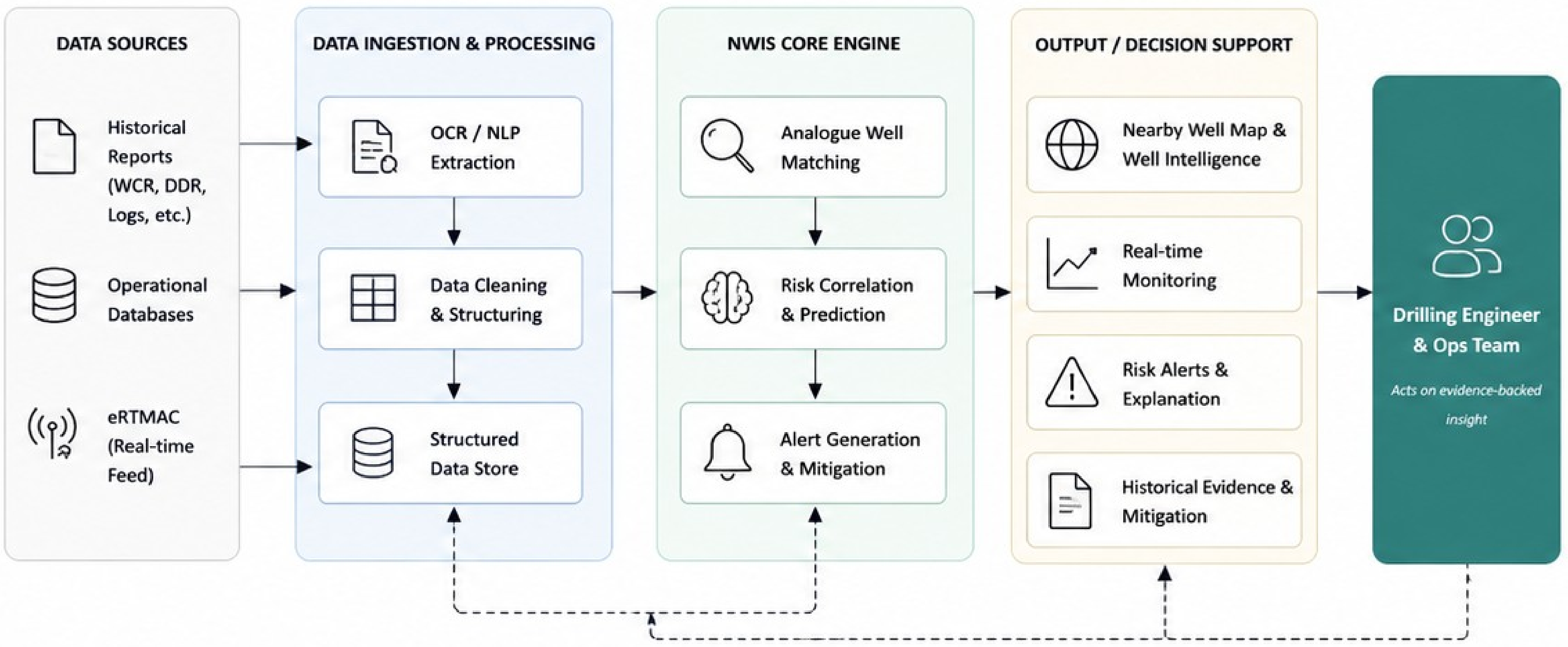
END-TO-END WORKFLOW

1. Open NWIS Dashboard
2. Select Active Well
3. View Nearby / Offset Wells
4. Rank Analogue Wells
5. Historical Well Intelligence
6. Live Telemetry Enters
7. Live + Historical Correlation
8. Risk Detected
9. Explainable Alert
10. Mitigation Recommendation
11. Engineer Reviews & Acts
12. New Knowledge Stored

FEEDBACK LOOP • engineer action → new knowledge → future wells

SYSTEM ARCHITECTURE



TECHNOLOGY STACK

OCR / NLP
Tesseract
spaCy
Transformers

Data & Storage
PostgreSQL
TimescaleDB
MinIO

ML / Similarity
scikit-learn
Sentence Embeddings
Rules Engine

Integration
REST APIs
WebSockets
eRTMAC Connector

Frontend
React
Leaflet / Mapbox
Chart.js

Others
Docker
FastAPI
Nginx

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FEASIBILITY AND VIABILITY

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FEASIBILITY — Why NWIS can be built and deployed today

- **Data Feasibility** — Uses existing drilling reports, well logs, trajectories, operational records and real-time telemetry.
- **Technical Feasibility** — OCR / NLP, geospatial analysis, ML and time-series processing are mature, deployable technologies.
- **Integration Feasibility** — Designed as a standalone intelligence layer alongside eRTMAC / WITSML / API–based systems.
- **Operational Feasibility** — Decision-support system — engineers remain in the loop for validating alerts and recommendations.
- **Secure Deployment** — Can be deployed on-premise / private infrastructure to protect sensitive drilling data.

SCALABILITY — How NWIS grows beyond the pilot

- **Modular Data Ingestion** — New wells and reports can be added without modifying the core system.
- **API-Based Integration** — Works alongside existing OIL platforms like eRTMAC using standard interfaces.
- **Distributed Processing** — Simultaneously processes live telemetry and historical data from multiple wells.
- **Independent Model Updates** — AI models can be retrained or upgraded without affecting other system components.
- **Plug-and-Play Expansion** — Deploy to new oil fields by simply connecting their data sources.

KEY CHALLENGES — AND HOW WE SOLVE THEM

DOMAIN	CHALLENGE	OUR SOLUTION
TECHNICAL	**Proprietary eRTMAC access**	Telemetry Replay Engine for live-feed simulation
TECHNICAL	**Fragmented document formats**	OCR + NLP + structured knowledge extraction
DATA & OPERATIONAL	**Proprietary OIL data**	Volve dataset for prototype & validation
DATA & OPERATIONAL	**AI decision reliability**	Explainable, evidence-backed, human-in-the-loop alerts

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IMPACT, BENEFITS & SCALABILITY

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NWIS IMPACT
From data to institutional intelligence

Scattered Reports
Unstructured, siloed and hard to search

Structured Knowledge
Digitized and organized for easy access

Historical Intelligence
Patterns, risks and mitigations made discoverable

Real-Time Decisions
Contextual insights for safer, faster decisions

Institutional Memory
Knowledge that grows and empowers every future well

BENEFITS FOR STAKEHOLDERS
Who gains and how

FOR ENGINEERS
- Faster evidence-backed decisions
- Less time searching reports
- Better situational awareness with historical context

More confidence, less uncertainty

FOR ORGANIZATION
- Institutional memory retained
- Standardized drilling practices
- Knowledge survives workforce changes

Stronger knowledge base, long-term value

FOR OPERATIONS
- Earlier risk identification
- Reduced non-productive time potential
- Consistent operational decisions

Safer operations, higher efficiency

FOR DIGITAL TRANSFORMATION
- Converts unstructured reports into reusable knowledge
- Connects historical & live drilling intelligence
- Foundation for future autonomous drilling systems

Future-ready, AI-powered drilling intelligence

SUSTAINABILITY IMPACT
Building a smarter, more responsible drilling future

Reduces Drilling Failures
- Early risk detection
- Fewer formation-related issues and well failures

Optimizes Resource Use
- Better planning
- Lower material, mud and energy waste

Lowers Downtime
- Proactive insights
- Reduced non-productive time

Sustainable Operations
- Smarter, safer drilling practices
- Responsible resource utilization

SUPPORTING UN SDGs

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE

12 RESPONSIBLE CONSUMPTION AND PRODUCTION

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RESEARCH FOUNDATION & PRIOR ART

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INDUSTRY FOUNDATION

- **OIL eRTMAC** — Real-time drilling monitoring & analysis
- **WITSML / ETP** — Standards for wellsite data exchange

RESEARCH FOCUS

01 Offset Well Intelligence
– Analogue selection & contextual analysis

02 Drilling Risk Prediction
– ML-based, formation-specific risk

03 Real-Time Correlation
– Live telemetry + depth / formation match

DATA FOUNDATION

EQUINOR VOLVE DATASET

- ~40,000 files — Real North Sea field data
- Full cycle 2008–2016, released 2018
- Used for prototyping & validation
- Public proxy, not OIL proprietary data

EXISTING WORK → WHAT IT DOES → WHAT NWIS EXTENDS

EXISTING RESEARCH / TECHNOLOGY	WHAT IT DOES	WHAT NWIS EXTENDS
Automated Offset Well Analysis *SPE-201590-MS*	• Finds relevant offset wells • Informs new well design	**Contextual Analogue Engine** • Ranks by geology, formation & depth • Not proximity alone
Digital & Automated Offset Analysis *SPE-176693-MS*	• Digitises historical drilling experience • Structures it for retrieval	**Institutional Memory** • Problem → mitigation → outcome • Reusable on future wells
ML-Based Drilling Risk Prediction *IPTC-24701-MS*	• Predicts risk from offset data • Learns from historical behaviour	**Live + Historical Correlation** • Anchors to ranked analogues • Uses active well's live depth
Real-Time Drilling Monitoring *OIL eRTMAC · WITSML / ETP*	• Streams live telemetry • Monitors the active well	**Proactive Risk Radar** • Adds depth-ahead alerts • Complements RT systems, not replaces